



The soil structural stability determined by the QuantiSlakeTest: Linkage with soil porosity, water-stable aggregate fractions and soil chemical properties

Toon van Dael^{*}, Quinten Verwerft, Erik Smolders[✉]

Division of Soil and Water Management, Department of Earth and Environmental Sciences, KU Leuven, Kasteelpark Arenberg 20 bus 2459, 3001 Leuven, Belgium

ARTICLE INFO

Handling Editor: Yvan Capowiez

Keywords:

Soil structure determination
Water-stable aggregates
QuantiSlakeTest
Water-retention curve

ABSTRACT

Recently, the QuantiSlakeTest (QST) was developed which records mass loss or gain of intact soil samples during submergence, but it is yet unclear how slaking relates quantitatively to soil structure. This study was set up to determine if the QST can be a quick and cost-effective alternative to established structure tests and can be linked to soil porosity, determined by the soil water retention curve, or to the water-stable aggregate (WSA, >250 μm) fraction obtained by wet sieving. This was performed in a comparative study with 22 soils collected in arable and pasture land with contrasting properties. The mass loss due to aggregate breakdown in the QST was smaller as soil organic carbon (SOC) content increased ($r = 0.59$), while SOC correlated weaker with the WSA fraction ($r = 0.45$) or with relative meso + macroporosity (> 30 μm ; $r = 0.53$). The WSA correlated strongly to oxalate extractable Fe (Fe_{ox}) in the soil ($r = 0.63$), conforming to earlier studies, but no such trend was found for QuantiSlake parameters, indicating that the QST captures a different aspect of soil structure than the wet sieving. The mass loss in the QuantiSlake decreased with increasing relative meso + macroporosity ($r = 0.68$), suggesting that soil resistance to aggregate breakdown is higher in soils with a large fraction of large pores than in soils with smaller average pore sizes. In a subset of 10 soils, root mass within each Kopecky ring was recorded. It showed that the QST and relative microporosity strongly correlated with the root mass in the sample ($r = 0.77$ and 0.71 , respectively), showing the importance of roots on soil structure. The QuantiSlake parameters have a much larger variability among sampling replicates ($\text{CV} = 75\text{--}100\%$) than the parameters of the two other methods. However, the QuantiSlake parameters differed much more among soils, leading to similar statistical power as wet sieving and the water retention curves. It is concluded that the QST is a useful additional index of soil structure in undisturbed soils. It even outperforms the wet sieving method to indicate mesoporosity and is thus better suited for studying, e.g., preferential flow.

1. Introduction

Soil structure is defined as the size, shape, and arrangement of individual soil particles, the consolidation of these particles into aggregates, the subsequent organisation of these aggregates, and the pore space created due to this aggregation (Bronick & Lal, 2005; Oades, 1993; The Royal Society, 2020). Pore size distribution, which is an aspect of soil structure, affects soil moisture retention as a function of water tension. Soil structure also affects the transport of solutes in soil. The preferential flow of solutes can bypass reactive surfaces yielding non-equilibrium transport (Jarvis, 2007). Along the same lines, it is well established that plant roots do not access all pores or pore size classes;

solutes migrate to the roots via diffusion and mass flow (Barber, 1995). The effects of soil structure on plant nutrient bioavailability are much less explored than in the soil physical field. Indeed, almost all traditional methods for diagnosing soil fertility disrupt the soil structure for pragmatic reasons. Currently, there is a knowledge gap in this field of study. To better understand the role of soil structure on plant nutrient availability, soil structure must be operationally well-defined. As opposed to several soil chemical properties, soil structure is affected by sieving and should be preferentially determined in undisturbed conditions. So there is a need for readily quantifying soil structure in intact samples.

Soil structure depends on the way aggregates are formed from the soil colloids. Aggregate formation in soil results from complex

^{*} Corresponding author.

E-mail address: toon.vandael@kuleuven.be (T. van Dael).

<https://doi.org/10.1016/j.geoderma.2025.117281>

Received 2 December 2024; Received in revised form 24 March 2025; Accepted 30 March 2025

Available online 18 April 2025

0016-7061/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

interactions between various physical, chemical, and biological processes controlled by soil properties. According to the aggregate hierarchy theory, macroaggregates ($>250\ \mu\text{m}$) are formed from combining earlier-formed microaggregates ($<250\ \mu\text{m}$; Oades & Waters, 1991). The macroaggregates are, on average, less stable than microaggregates and will easily break apart under external forces such as tillage (Kleber et al., 2007; Oades, 1993; Totsche et al., 2018). The stability of these macroaggregates is an essential aspect of soil structure since unstable macroaggregates will decrease the mesoporosity and, thus, the preferential flow of solutes (Zhang et al., 2021).

The aggregate breakdown is often attributed to four different mechanisms: slaking, breakdown by differential swelling of clays, mechanical breakdown by impact and physicochemical dispersion due to osmotic stress (Le Bissonnais, 1996). Which mechanism is the most important depends on soil properties such as clay content but also on vegetation cover and root density (Agassi et al., 1985; H. Xiao et al., 2018). Soil pH strongly affects aggregate stability because it controls the concentration of multivalent cations in the soil. At low pH, humic acids bind better to metal oxyhydroxides and organic and mineral particles coagulate more strongly due to a decrease in electrostatic repulsive forces, thus improving the aggregate stability (Regelink et al., 2015). By contrast, the high pore water Ca^{2+} concentration in calcareous soils is essential for aggregate stability in those soils as it reduces the thickness of the double layer between aggregates, thereby decreasing repulsive forces. Hence, a positive relationship is observed between the pore water Ca^{2+} concentration and aggregate stability (Safar & Whalen, 2023).

There are various methods to determine soil structure, the preferred method depends on the process that is being studied. The aggregate size distribution is the most widely used method, in which a wet sieving procedure is used to determine the water-stable aggregate fraction (WSA). During wet sieving, air-dried samples are rewetted and immersed in water while placed on sieves of different sizes, with often preset oscillations shaking the sieve. The proportion of fragments $> 250\ \mu\text{m}$ constitutes the water-stable aggregates (Panagea et al., 2022; Rabot et al., 2018; Regelink et al., 2015). The WSA fraction has been well-correlated with soil properties important for aggregate formation and stability. Increasing Fe oxyhydroxide and SOC content correlated with increasing WSA fractions in a wide range of soil samples (Regelink et al., 2015). The wet-sieving methods are most commonly destructive and start with samples that are dry-sieved over 8 mm before the wet-sieving procedure, thereby already changing some structural properties present in soils in the field. Determining soil structure in undisturbed conditions, is commonly based on imaging techniques. Visual micromorphology in 2D was a central point of attention in the early stages of soil science (Kubiena, 1938). Advanced 3D imaging techniques with CT tomography are often proposed and used to quantify pore size distribution, root architecture, soil phase classification and solute transport in soil with high resolution, but these are expensive and time-consuming methods (Ghosh et al., 2023; Papadopoulos et al., 2009).

The soil water retention as a function of water tension determines the soil porosity and, hence, can be classified as an indicator of soil structure (Bleam, 2017; Regelink et al., 2015). Other functional tests for soil structure are the soil dispersion tests (Craze & Hamilton, 1991; Standards Association of Australia, 1980; Richie, 1963) or tests of the resistance to soil slaking (USDA, 2008; Vanwindekens & Hardy, 2023). The soil dispersion test indicates the structural stability of the soil. Sodic soils are the most sensitive to this phenomenon since sodium (Na^+) interferes with the ability of divalent cations to bind and stabilise clay particles (Bless et al., 2022). Slake tests demonstrate the stability of soil aggregates in water. When a topsoil sample is placed into water, the water starts filling the pores and displaces air. If the large pores within the soil are stable, water can move into the soil without causing the aggregate to break apart ("slake"). Such tests can indicate, for example, that reduced tillage or increasing organic matter applications to soil lowers soil aggregate breakdown (Vanwindekens & Hardy, 2023).

The recently developed QuantiSlakeTest (QST, Vanwindekens &

Table 1

location, coordinates (WGS 84), WRB soil type and land use of each of the 22 soils used in this study.

Soil number	Location	Coordinates	Land use	Soil type
1	Duisburg	50°48'40.1" N 4°33'52.2" E	Cropland	Luvisol
2	Steenokkerzeel	50°57'00.3" N 4°29'27.9" E	Cropland	Retisol
3	Vorselaar	51°14'16.6" N 4°44'41.2" E	Cropland	Podzol
4	Kasterlee	51°14'54.5" N 4°58'57.5" E	Cropland	Podzol
5	Hageven	51°14'45.7" N 5°24'30.0" E	Pasture	Podzol
6	Bocholt 1	51°10'53.4" N 5°37'34.6" E	Pasture	Podzol
7	Bocholt 2	51°10'43.5" N 5°37'46.1" E	Pasture	Histosol
8	Schaffen	51°00'12.5" N 5°05'36.7" E	Pasture	Cambisol
9	Wilsele 2	50°54'28.1" N 4°41'55.7" E	Cropland	Luvisol
10	Wilsele 1	50°54'35.3" N 4°41'37.4" E	Cropland	Luvisol
11	Lovenjoel	50°51'15.8" N 4°46'11.8" E	Cropland	Luvisol
12	Ter Munck	50°52'44.5" N 4°39'26.4" E	Cropland	Luvisol
13	Pair-et-grandrupt	48°16'26.0" N 7°12'27.5" E	Pasture	Fluvisol
14	Bettancourt	48°38'57.8" N 4°56'47.0" E	Pasture	Luvisol
15	Fontenailles	48°32'55.7" N 2°55'54.5" E	Pasture	Luvisol
16	Ervauville	48°5'27.2" N 2°59'57.8" E	Cropland	Leptosol
17	Sours	48°26'19.3" N 1°35'22.9" E	Cropland	Cambisol
18	Le Mesnil-Simon	48°53'25.8" N 1°30'58.3" E	Cropland	Cambisol
19	Touffrevil	49°20'45.2" N 1°26'33.7" E	Cropland	Colluviosol
20	Vaudesson	49°28'35.0" N 3°29'22.9" E	Cropland	Cambisol
21	Hirson	49°57'29.9" N 4°6'20.2" E	Pasture	Cambisol
22	Hélesmes	50°21'47.2" N 3°21'51.8" E	Cropland	Leptosol

Hardy, 2023) is a quantitative, high-throughput method that can be used on intact soils sampled directly in the field, including plant roots. The method involves submerging an air-dried, intact soil core and following up the mass change over time. Soils with more stable aggregates are expected to retain more mass. The fact that intact samples are used offers specific advantages compared to other tests, such as wet sieving, because plant roots and preferential flow paths, that affect aggregate stability, are not removed before the analysis (Totsche et al., 2018). Vanwindekens & Hardy (2023) already showed for different soil management practices that the QST correlates well with the mean weight diameter (MWD) and percentage of macro aggregates in the fast-wetting test of Le Bissonnais and with MWD of the slow-wetting test. It is unclear if that method can discriminate soil structure among soils with more varying soil properties than based on soil management.

The main goal of this study was to confirm if the QST is a viable alternative to characterise soil structural stability in comparison to more time- and labour-intensive methods such as wet sieving. Therefore, this study was set up to determine if the QST can be linked to soil porosity, the WSA fraction, and soil chemical properties that are known to affect soil structure (Regelink et al., 2015). At the same time, this study may specify an alternative and convenient method for determining soil structure on intact soil samples. Practical aspects of the test, such as variability and test time, are considered.

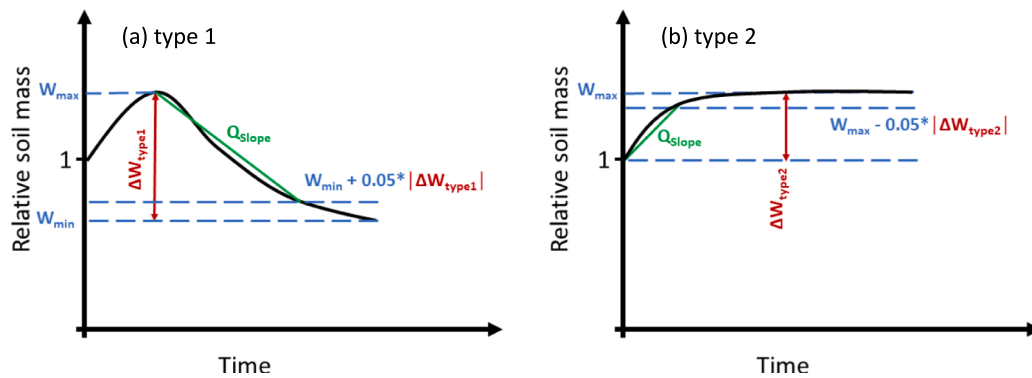


Fig. 1. Schematic representation of the two types of mass evolution when using the QST. Part (a) shows an unstable soil that loses much of its mass during immersion and the parameters used to quantify this effect. Part (b) shows the same for a soil with stable aggregates. All soil masses are rescaled to the mass at the start of submergence.

2. Material and methods

2.1. Study sites and soil sampling

In total, 22 agricultural topsoils were sampled in cropland or grassland in Belgium and France in the summer of 2023. Soil samples had contrasting properties that were expected to affect soil structure, i.e., a wide range in soil pH, metal oxyhydroxides, and organic matter contents. The WRB soil types of the samples include Luvisols, Retisols, Podzols, Cambisols, Leptosols, and Fluvisols. The location of each sampling point is shown in Table 1 and Fig. S1.

At each sampling location, 3 kg of fresh bulk soil was sampled from the 0–15 cm layer after removing the vegetation (if present). One bulk sample consisted of 4 subsamples taken in an area of 10 m². Besides bulk samples, six undisturbed samples were taken from the top 5 cm using Kopecky rings with a 53 mm diameter and 45.3 mm depth. The bulk soil was sieved over 8 mm and air-dried before analysis of the sieved fraction. Three intact samples were air-dried in the Kopecky rings. The other three intact samples were stored cold and used to determine the soil moisture retention (pF curves) within 1 day after sampling; thus, they were not dried.

2.2. Soil physicochemical characterisation

Total carbon (%TC) and organic carbon (%OC) contents were determined on oven-dried samples at 105 °C with an elemental analyser (Carlo Erba EA1108) in tin capsules with or without the removal of carbonates with concentrated HCl. The pH was measured in an extract of 0.01 M CaCl₂, with a solid:liquid ratio of 1:10. The concentrations of amorphous Al and Fe oxyhydroxides were determined with oxalate extraction on soil samples dried at 45 °C according to Schwertmann (1991), followed by measurement with ICP-OES (ICP-OES Thermo Scientific iCAP 7000 series). The effective CEC of the soil samples was determined by extraction with a cobalt hexamine chloride solution (Cohex, ISO 23470, 2007) on soil samples dried at 45 °C followed by Co measurement with ICP-MS (ICP-MS Agilent 7700x, Agilent Technologies). Particle size analysis was conducted using laser beam diffraction (LS 13 320 Laser Diffraction Particle Size Analyser, Beckman Coulter, (ISO 13320:2020, Particle Size Analysis — Laser Diffraction Methods, 2020)). First, air-dried soil samples were homogenised and rewetted with demineralised water. Then, the inorganic carbon was removed by adding 10 % HCl, followed by organic carbon removal with 35 % H₂O₂. Afterwards, the samples were treated with an ultrasound rod to disperse the clay particles. Finally, the samples were analysed with laser diffraction. All these soil analyses included internal soil reference materials and analytical replicates to ensure accuracy and precision. All results are reported on a dry weight (105 °C) basis.

2.3. Soil structure analysis

Three methods were used as possible proxies for soil structure: the water-stable aggregate (WSA) fraction, the water-retention curve (pF curve) and the QST. The WSA is an established method for characterizing aggregate stability; Porosity can be approached using the water retention curve (Panagea et al., 2022; Regelink et al., 2015), while the QuantiSlakeTest is a new method for estimating structural stability on intact soil samples developed by (Vanwindekens & Hardy, 2023).

Wet sieving was used to determine the WSA fraction according to Elliott (1986). For the wet sieving method, an < 8 mm sieved, dried (105 °C) soil sample of 100 g was placed on a sieve and submerged in deionised water while being shaken for two minutes with 25 rotations per minute, according to Panagea et al. (2022). The soil fraction remaining on the sieve was collected in an aluminium tray and dried in an oven at 105 °C. The soil that passed the first sieve was then transferred to the next, smaller sieve, and the process was repeated. Three sieves were used (2 mm, 250 µm and 53 µm). This way, four fractions were obtained: > 2 mm, 250 µm – 2 mm, 53 µm – 250 µm, and < 53 µm. The fractions that remained on each sieve were oven-dried at 105 °C and weighed to obtain a mass fraction distribution. The fraction < 53 µm was not weighed but calculated from the mass balance of the other fractions and the initial total soil mass. The aggregates > 250 µm that remain stable during the wet sieving procedure are the WSA fraction. All analyses were executed in duplicate.

The soil water retention curves were determined using the sand bath and pressure plate methods (Buckingham E., 1907). It used intact soil samples collected in triplicate with 100 cm³ Kopecky rings. The soils within the rings were saturated with water, and the evolution of the moisture content was determined by weighing the rings and determining the moisture content at different pressures: pF 0, pF 0.5, pF 1.0, pF 1.5, pF 1.8, pF 2.0, pF 2.3, pF 2.8, pF 3.4 and pF 4.2. The rings were left standing for one week at each pressure to ensure equilibration of the water content in the soil. To measure the highest pF values (3.4 and 4.2), the samples were transferred to smaller rings, rehydrated, and placed in a pressure vessel to achieve the necessary pressures. Afterwards, the samples were oven-dried at 105 °C and their dry masses were measured. The soil water content θ_x at each pF value x and the bulk density was determined from these masses. The sample volume of 100 cm³ was corrected for the volume of rocks inside the Kopecky rings, by removing the rocks after the procedure and accurately measuring their volume by submerging them in a graduated cylinder. The total porosity of each soil equals the water content at saturation (θ_0), the microporosity is defined as the pores filled at pF 2.0 ($\theta_{2.0}$), which corresponds with pore sizes < 30 µm, according to the Young-Laplace equation, and the meso + macroporosity are the pores that drain between pF 0 and pF 2.0 ($\theta_0 - \theta_{2.0}$, pores > 30 µm). This definition pools both the macro- and mesopores into one group, as it was expected that both fractions would have a

Table 2

Summary statistics of the distribution of soil characteristics for all soil samples. the first and third quartiles correspond to the 25th and 75th percentiles, respectively. The measured characteristics include the cation exchange capacity (CEC), pH in CaCl₂ solution, oxalate-extractable iron and aluminium (Fe_{ox}, Al_{ox}), and particle size analysis.

	CEC	pH	OC	Fe _{ox}	Al _{ox}	Clay fraction	Silt Fraction	Sand fraction
	(cmolc kg ⁻¹)	(–)	(%)	(mg kg ⁻¹)	(mg kg ⁻¹)	(%)	(%)	(%)
Max	16.76	7.24	4.13	6200	2200	51	78	80
Q3	12.89	6.64	2.14	3000	860	10	75	60
Median	7.62	6.20	1.64	2400	590	6.4	64	23
Q1	6.29	5.30	1.28	1700	520	3.5	29	17
min	2.36	4.00	0.80	1000	270	0.1	16	10

similar relation to soil structure. Mesopores range between 30–75 µm diameter and macropores are > 75 µm (Weil & Brady, (2016)). We defined the relative micro- and mesoporosity (Eqs. (1) and (2)) as the fractions of micro-, and meso + macroporosity relative to the total porosity (water content at pF = 0).

$$\text{Relative microporosity} = \frac{\theta_{2.0} - \theta_{4.2}}{\theta_0} \quad (1)$$

$$\text{Relative meso + macroporosity} = \frac{\theta_0 - \theta_{2.0}}{\theta_0} \quad (2)$$

The QST tested the structural stability of soil under submerged conditions and was performed according to (Vanwindekens & Hardy, 2023). First, the air-dried soils were removed from the ring and placed in a basket with a mesh size of 8 mm. The basket was hung from a balance and then submerged in water. The mass of the sample was then continuously recorded with the balance (Sartorius, 0.01 g precision) connected to a PC. The soil was submerged until there were no mass changes for longer than a minute, which occurred between 10 and 25 min after submergence, depending on the soil. The aggregates, that were still present afterwards, were assumed to be resistant to slaking. For the ten French soils, the remaining soil in the bucket was washed for ten sampling sites, and the roots were collected and air-dried before weighing them. The mass versus time curves follow one of two types depicted in Fig. 1. Type 1 is the soil for which an initial mass increase is observed due to the rewetting of the sample, followed by a decrease due to aggregate breakdown. Type 2 is the soil for which only a mass increase is observed because aggregation is so strong that aggregate breakdown does not occur. Curve fitting was used to analyse data. All masses are rescaled to the initial soil mass W at the start of submerging the air-dried soils. For Type 1, the slope of weight loss (negative slope) is defined as the rate of mass loss between the maximum mass and the point 5 % above the minimum mass ($Q_{\text{slope}, \text{type1}}$):

$$Q_{\text{slope}, \text{type1}} = \frac{(W_{\min} + 0.05 \cdot |\Delta W_{\text{type1}}|) - W_{\max}}{t_{\min+0.05} - t_{\max}} \quad (3)$$

With $Q_{\text{slope}, \text{type1}}$ the rate of mass loss for type 1 soils, $W_{\min}$ is the minimum mass of the soil during slaking, $W_{\max}$ is the maximum mass and ΔW_{type1} the difference between $W_{\max}$ and $W_{\min}$, $t_{\max}$ is the time at which the maximum mass is reached and $t_{\min+0.05}$ the time at which the mass is 5 % above the minimum mass. Note that for Type 1 soils, ΔW is defined to be negative.

For Type 2, the slope of the mass gain (positive slope, $Q_{\text{slope}, \text{type2}}$) was defined as

$$Q_{\text{slope}, \text{type2}} = \frac{(W_{\max} - 0.05 \cdot |\Delta W_{\text{type2}}|) - W_{\min}}{t_{\max-0.05} - t_{\min}} \quad (4)$$

With $t_{\max-0.05}$, the time at which the mass is 5 % below the maximum mass and ΔW_{type2} , the difference between $W_{\min}$ and $W_{\max}$; note that for type 2 soils, ΔW is positive. The other variables are defined the same as for type 1 soils. From Eqs. (3) and (4), it is clear that soils with a strongly negative slope are considered less stable than soils with a positive or slightly negative slope. This way, both Type 1 and Type 2 samples can be compared simultaneously. The slope that results from Eqs. (3) and (4)

relate the the equilibrium of two processes: mass loss due to aggregate breakdown and mass gain due to water uptake within the soil core. In type 1 soils, the mass loss rate overtakes the water uptake rate at the maximum of the curve, while in type 2 soils, the water uptake rate remains dominant. It must be noted that the slope for type 1 soils is not calculated starting from $t = 0$ as for type 2 soils. This is due to the fact that all correlations between Q_{slope} and soil properties were weaker when this method was applied. The same is true for the second parameter used from the QST, ΔW . Strongly negative values for ΔW indicate significant mass loss due to aggregate breakdown, hence a less stable structure, while positive or slightly negative values indicate a stable soil structure.

Note that the original development of QST used several other indicators of the mass-time diagram. Here, we restricted that to the two described above. This was done for simplicity and because the multiple slope parameters originally described by Vanwindekens & Hardy (2023) all showed similar correlations with soil properties and the three tests of aggregate stability of Le Bissonnais (1996).

2.4. Statistical analyses

Statistical analyses were executed using JMP® Pro version 17.0.0. Before analysis, the distribution of the different soil characteristics was checked for normality by performing a skewness (symmetry of the distribution) and Kurtosis test (flatness of distribution). A normal distribution was assumed if both values were within a range of –2 and 2 (Brown, 2011), which was the case for all parameters considered. Single correlation analyses between soil properties and soil structure parameters were done by calculating Pearson's correlation coefficient. The soil properties included were the pH, Fe_{ox}, %OC, soil porosity and root mass. The coefficient of variation (CV, %) was calculated for the five structure indicators. The CV is defined as the standard deviation divided by the absolute value of the mean multiplied by 100. The CV of the replicates of each sampling location was calculated as well as the CV for the mean parameter values among all soils. The ratio of both these CV values can be considered a measure of the variability of an analysis method relative to its variability among different soils, i.e. the discriminatory power of a certain parameter or indicator.

3. Results and discussion

3.1. Soil characteristics

The soils were characterised for their organic carbon content, oxalate extractable Fe, Al, CEC, pH, and particle size. Table 2 shows the median values and ranges. The complete characterisation of each soil can be found in the supporting information (Table S2).

The range of the measured properties shows that the current dataset is an extension of the range of soil types, soil properties and land covers compared to Vanwindekens and Hardy (2023). This indicates that the soil sample set used here can be used as an approximation for an average Western European soil. However, a more extensive dataset is necessary to validate these findings.

Table 3
Correlations among the soil structure parameters: the water-stable aggregate (WSA) fraction, all wet sieving fractions, the relative micro- and meso + macroporosity, and the QST parameters (top half of the table), and between the structure tests and soil properties shown in Fig. S2 and Fig. 2 (bottom half of the table). Values in blue are significant positive correlations, and values in red are significant negative correlations; values in grey are not significant at $p = 0.05$.

		Wet Sieving					Water retention curve		Quantislate	
		WSA fraction (g g ⁻¹)	Fraction >2 mm (g g ⁻¹)	Fraction 250 µm – 2 mm (g g ⁻¹)	Fraction 53 µm – 250 µm (g g ⁻¹)	Fraction <53 µm (g g ⁻¹)	Rel. Microporosity (cm ³ cm ⁻³)	Rel. Meso + macroporosity (cm ³ cm ⁻³)	ΔW	Q _{slope} (s ⁻¹)
WSA fraction	g g ⁻¹	1.00								
Fract. > 2 mm	g g ⁻¹	0.42	1.00							
Fract. 250 µm – 2 mm	g g ⁻¹	0.36	-0.70	1.00						
Fract. 53 µm – 250 µm	g g ⁻¹	-0.91	-0.40	-0.31	1.00					
Fract. < 53 µm	g g ⁻¹	-0.37	-0.12	-0.17	-0.04	1.00				
Rel. Microporosity	cm ³ cm ⁻³	-0.30	-0.25	0.02	0.23	0.22	1.00			
Rel. Meso + macroporosity	cm ³ cm ⁻³	0.38	0.02	0.28	-0.26	-0.35	-0.88	1.00		
ΔW	–	0.12	0.23	-0.14	0.02	-0.34	-0.65	0.68	1.00	
Q _{slope}	s ⁻¹	0.20	0.12	0.03	-0.16	-0.12	-0.34	0.36	0.61	1.00
%OC	%	0.45	0.02	0.34	-0.25	-0.52	-0.31	0.53	0.59	0.44
Fe _{ox}	mg kg ⁻¹	0.63	0.33	0.16	-0.52	-0.36	0.24	-0.17	0.24	-0.27
pH	–	-0.31	0.32	-0.57	0.10	0.51	0.02	-0.36	0.31	-0.25
Total porosity	cm ³ cm ⁻³	0.68	0.35	0.18	-0.62	-0.25	-0.62	0.70	0.53	0.25
Microporosity	cm ³ cm ⁻³	0.02	-0.10	0.11	-0.07	0.13	0.89	-0.70	-0.53	-0.29
Meso + macroporosity	cm ³ cm ⁻³	0.45	0.06	0.29	-0.33	-0.35	-0.87	0.99	0.68	0.36
Root mass	g ring ⁻¹	0.11	0.16	-0.10	-0.10	-0.07	-0.71	0.48	0.77	0.80

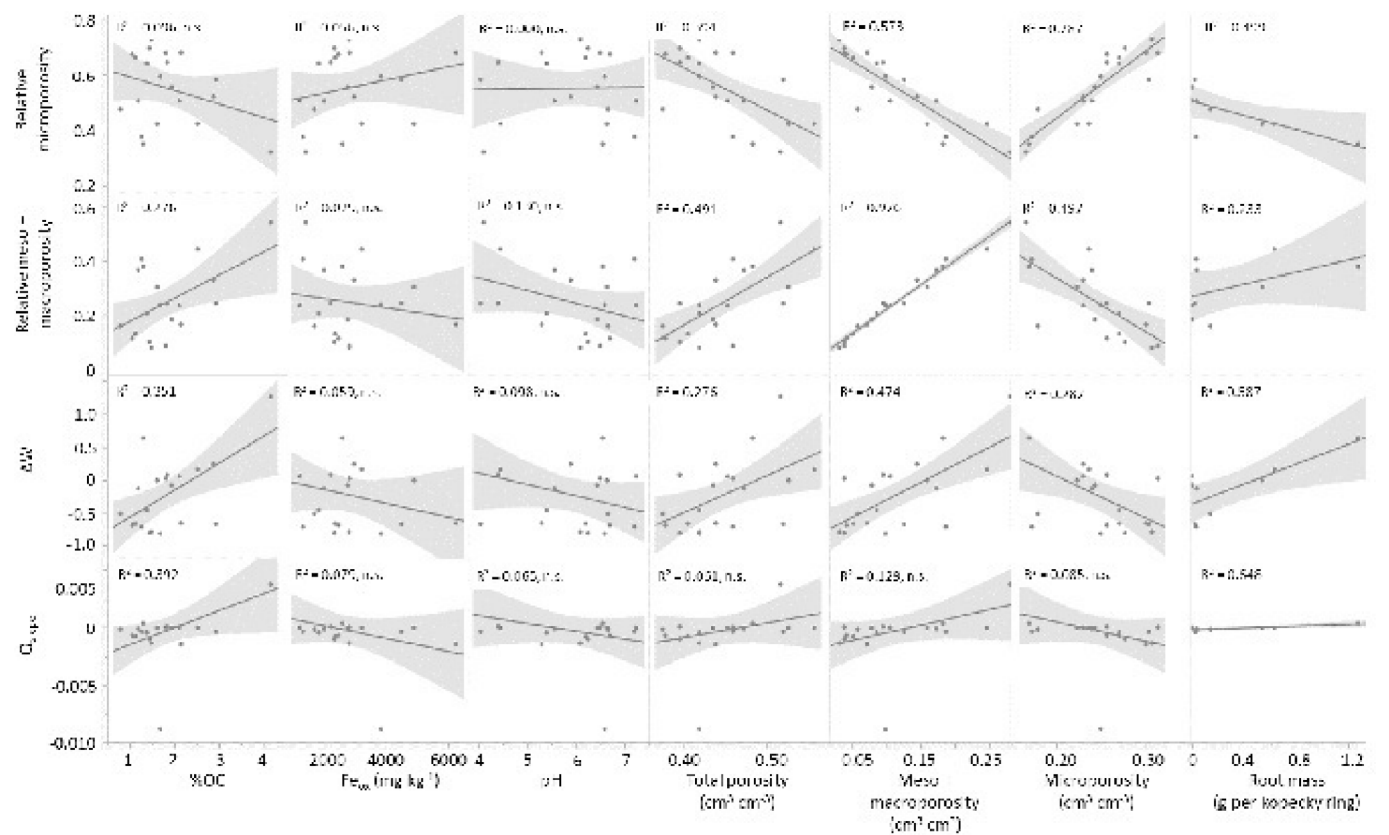


Fig. 2. The parameters for the water retention curve and the Quantislate, i.e., the slope and the difference between minimum and maximum mass (Q_{slope} and ΔW), are shown as a function of soil chemical parameters of interest: the organic carbon content, oxalate extractable Fe, pH, porosity, and root mass in the intact samples. A linear regression line is plotted to guide the eye, and the R² is shown for each graph. The gray areas indicate the 95% confidence intervals for each figure.

3.2. Soil structure parameters

Table S2 shows the values for all indicators considered for the three soil structure tests. For the wet-sieving method, the WSA fraction, as well as each sieving fraction, was considered. The micro and meso + macroporosity relative to the total porosity were considered for the water retention curve. The QST parameters consisted of the slope of the curve and the difference between maximum and minimum relative mass.

The QST data shows that among the 22 soils, 13 were of type 1, i.e. with mass loss due to aggregate breakdown, and nine were of type 2, i.e. relative mass increased due to water uptake, but no mass was lost due to aggregate breakdown. Some soils ($n = 2$) of type 1 did not exhibit an initial net water uptake and started to slake directly after submergence.

The data from the wet sieving method shows that the WSA fraction ranged from 0.48 to 0.93, with an average value of 0.69. The mean of individual fractions were 0.15 (>2 mm), 0.54 ($250\ \mu\text{m} - 2$ mm), 0.23 ($53\ \mu\text{m} - 250\ \mu\text{m}$) and 0.08 ($<53\ \mu\text{m}$).

The porosity data show that the micropores were the largest of the total pore fraction (mean 0.55), while the meso + macropore fraction was, on average 0.25. the remainder of the pores consisted of ultramicropores ($<0.1\ \mu\text{m}$, $> \text{pF } 4.2$). The water in this pore fraction is entirely inaccessible to plants. the ultramicropore fraction was not included in the micropore fraction in this study because the correlations of the microporosity with all other parameters became weaker when the ultramicroporosity was included. The water in this fraction is mainly present between individual silt and clay particles and is thus mainly dependent on the silt and clay fractions (data not shown).

3.3. Soil properties affecting the QST

Table 3 and Fig. 2 show the univariate correlation coefficients among the different structure parameters from this study, as well as the correlation coefficients for relevant soil characteristics proven to affect soil structure (Regelink et al., 2015). The QST parameters show a clear correlation with %OC: increasing %OC increased the slope (Q_{slope}) and decreased the difference in mass between maximal and final mass after slaking (ΔW), i.e. large %OC values lead to a more stable soil core. This coincides well with the findings of (Vanwindekens & Hardy, 2023). A strong correlation exists between the porosity and the ΔW parameter, with large total porosity and meso + macroporosity indicating a more stable soil structure. The relationship with the total and relative meso + macroporosity is especially strong ($r = 0.68$ for both). For microporosity, the relation was reversed. The meso + macroporosity mainly consists of pores between the water-stable aggregates and even larger aggregates > 8 mm, while the micropores are found within these aggregates (Panagea et al., 2022). Fig. S3 shows that The QST parameters are not affected by the clay content of the soil. This is in contrast with the study by Vanwindekens & Hardy, (2023).

The soil Fe_{ox} content did not affect the QST parameters. In previous studies, the Fe_{ox} fraction was shown as an essential parameter to predict structural stability, i.e. the WSA as corroborated also here (Table 3). This indicates that the water retention curve and QST capture a different aspect of soil structure than the WSA. A possible explanation is that Fe_{ox} is mainly vital for forming aggregates smaller than 8 mm (the grid size of the QST) when coagulating around an organic core material (Totsche et al., 2018). This theory is strengthened by the strong positive correlation of Fe_{ox} with the WSA fraction ($r = 0.63$), which consists of all aggregates between $250\ \mu\text{m}$ and 8 mm.

The QST and the water retention curve were used with intact samples, and the importance of the root mass is immediately apparent. The correlation coefficient of root mass is the highest for both parameters of the QST and also highly significant for the water retention curve parameters ($p < 0.001$). The fact that this correlation is absent for the wet sieving method shows that this method might not encompass one of the most critical factors affecting a soil structure since the roots are removed

during sample pretreatment (Panagea et al., 2022). This can be especially important with regard to mesoporosity and, by extension, macroaggregation (Zhang et al., 2021). This is where the roots play a vital role as binding materials and by forming root channels that form large pores. Xiao et al. (2020) found the same positive effect of roots on aggregate stability. They noticed an increase in the mean weight diameter of the soil aggregates from 1.15 to 1.3 mm when the specific root length increased from 0.25 to 2.2 m/g, i.e., when the roots became thinner. Root length density and root biomass were significantly higher (82 % and 67 %, respectively) around preferential flow pathways within the soil profile compared to the bulk soil (Zhang et al., 2015). This again stresses the importance of roots for soil structure, especially in determining the large pores and preferential flow. Note that the root masses were only measured in the French soils ($n = 10$), so a more extensive dataset is necessary to validate these results.

3.4. Soil properties affecting wet sieving fractions

Increasing OC and Fe_{ox} contents significantly increased the WSA fraction. Neither pH nor soil particle size fractions significantly affected the aggregate stability in our samples. Earlier studies also found that soil structure strongly correlated with %OC (Benjamin et al., 2008; de Dios Herrero et al., 2020; Johannes et al., 2023). Benjamin et al. (2008) found a positive correlation between the stable macroaggregates in soil and % OC, while de Dios Herrero et al. (2020) observed the same positive effect for the total porosity. This effect is attributed to the interaction between OC and clay particles, stabilising microaggregates. Johannes et al. (2023) found that OC% was a better explanatory variable for a high water holding capacity than was clay content, again highlighting the importance of %OC. The glueing effect of Fe oxyhydroxides found in earlier studies (Totsche et al., 2018) was confirmed in this study. Duiker et al. (2003) found that Fe_{ox} concentrations were strongly correlated with the WSA fraction. In their research, an increase of 1000 mg Fe_{ox} kg^{-1} increased the WSA fraction by 20 %. In contrast to earlier studies (Johannes et al., 2023; Newman, 1984), our results did not show significant effects of particle size on aggregate stability (Fig. S3). Only the clay fraction slightly affected the WSA fraction, but this trend is not significant even though the soils in the dataset encompass a wide range of soil textures (clay content ranged between 0.1 – 51 %). This lack of effect might be explained by the fact that the clay surfaces are not the main sorption surface for OC in our study sites. Instead, results indicate that Fe_{ox} is the main adsorption surface, shown by its strong correlation with the WSA (Table 3) and the organic carbon content of the soil. Additionally, the effect of clay on structure depends on the clay mineralogy (Six et al., 2000). Mineralogy was not measured in this study, but if non-swelling clays such as illite dominate the clay fraction, the effect on soil structure is expected to be much less pronounced compared to when swelling clays are present, such as montmorillonite or smectite (Brubaker et al., 1992).

A well-structured soil typically exhibits a good water balance, reflected by a strong correlation with porosity. The WSA mainly correlated to total and meso + macroporosity in this study, but there was no relationship with microporosity. This contrasts with results found by Regelink et al. (2015) and is somewhat surprising, considering that microporosity is most closely related to the available water holding capacity, i.e., the water available for plants (water present in pores $< 30\ \mu\text{m}$). Finally, the effect of pH on the WSA fraction, found in many studies (Regelink et al., 2015; Stătescu et al., 2013), is lacking here. In theory, a decreasing pH will strengthen the interaction between OC and Fe_{ox} , leading to more stable aggregates (Feng et al., 2005; Weng et al., 2008). Even though the WSA fraction does not follow this trend in this set of soils, the strong negative correlation between pH and the $250\ \mu\text{m} - 2$ mm fraction and the positive correlation between pH and the fraction $< 53\ \mu\text{m}$ indicate that lower pH does lead to more stable aggregates in our study sites. Fig. S3 shows a slight trend that decreasing pH increases structural stability. However, the relation is less pronounced than that

Table 4

Average of the coefficient of variation (CV, %) for the five structure parameters: WSA, relative micro- and meso + macroporosity, ΔW and Q_{slope} . The average CV of each parameter is calculated for the replicate samples of each location, $n = 2$ for WSA, and $n = 3$ for the other parameters. The inherent parameter variability is calculated as each parameter's CV among all soils. The final row shows the ratio of both these CVs.

	WSA fraction	Relative Microporosity	Relative Meso + macroporosity	ΔW	Q_{slope}
Average of the CV values of replicate measurements per soil	9	6	21	100	75
CV of the average values among all soils	20	23	52	240	1100
Ratio of the average replicate CV to the CV of averages among soils	0.45	0.26	0.40	0.42	0.07

found by Regelink et al. (2015) Regelink et al. (2015). A possible reason for this reduced effect is a lower content of active clays with a high CEC; the average CEC in this study was $8.8 \text{ cmol}_c \text{ kg}^{-1}$, with a maximum value of $16 \text{ cmol}_c \text{ kg}^{-1}$. Clay-dominated soils can reach CEC values of up to $40 \text{ cmol}_c \text{ kg}^{-1}$ (Kumaragamage et al., 2021).

3.5. Soil properties affecting the water retention curve

The soil properties that correlated with the water retention curve parameters are similar to those correlated with the QST. Increased %OC increased the relative meso + macroporosity, i.e. a more stable soil. Obviously, all fractions of soil porosity were strongly linked to the water retention curve parameters since this is a direct measurement of the pore size distribution. As for the QST, the root mass strongly affected the water retention of the soils. A large root mass corresponded with a large fraction of the porosity consisting of large pores.

3.6. Comparison of parameters of the QuantiSlakeTest, wet sieving test and the water retention curve

Table 3 shows that the relative meso + macroporosity and ΔW were strongly positively correlated, i.e. increasing fractions of mesoporosity increase soil resistance to aggregate breakdown. This indicates that both methods likely predict the same aspect of soil structure. In contrast, there are no significant correlations between the WSA fraction and the two other structure tests, even though they are, in part, determined by the same soil characteristics, i.e. %OC and total porosity. This suggests that WSA as a measure for soil structure is not directly interchangeable with the other methods. However, the soil water retention curve and QST, i.e. methods using intact samples, are better predictors for the soil porosity (especially the large soil pores). This is likely due to roots in the intact samples, which are essential for macroaggregation and mesoporosity (Xiao et al., 2020). This indicates that the water retention curve and QST could act as viable alternatives for the wet sieving method as a proxy for soil structure when the research focuses mainly on (meso) porosity and preferential flow. If the study focuses more on the soil properties and the mechanisms behind aggregation, the WSA fraction is a better indicator of soil structure.

3.7. Practical aspects

The wet sieving procedure is quite labour-intensive as each sample requires the collection of three soil fractions from different sieves, which require meticulous cleaning during the sample collection and between samples (Panagea et al., 2022). Completely measuring one sample may take about one hour. The water retention curve requires the equilibration of Kopecky rings at nine different pressures. The rings need one week of equilibration time per pressure, meaning a total analysis time of nine weeks. The QST, in contrast, can be placed on a balance for about 15–20 min until the mass stabilises (Vanwindekens & Hardy, 2023), making it the most practically applicable method in terms of labour and time.

Next to the research focus and applicability, another critical factor to consider is the variability of each method. The QST and water retention curve use small volumes of intact soil (100 cm^3) from a single point in a sampling location. In contrast, the wet sieving method uses a sieved

sample consisting of multiple subsamples. This means the QST and water retention curve are probably more susceptible to heterogeneities in the field. Hence, assessing the variability among replicates of a single soil is critical. Table 4 shows that the QST (ΔW and Q_{slope}) had by far the largest replicate CVs and was thus the most sensitive to local differences within one sampling location. However, as the variation of the QST parameters among soils is 5 to 10 times greater than that of the other methods, it remains possible to statistically detect differences among soils without having to increase the number of replicates, even when the replicate CVs within a single soil are 5 to 15 times greater than those of the other methods (Table 4). This is reflected by the ratio of both CVs, which was in the same order of magnitude for all parameters, indicating a similar discriminatory power (statistical power) to detect differences among soils even with only three replicate samples. In fact, Q_{slope} even has the lowest ratio, hence the strongest discriminatory power.

4. Conclusion

The results in this study confirm that %OC and Fe_{ox} correlate strongly positively to the WSA fraction of soils. However, due to the pretreatment necessary for the wet sieving method, an essential factor controlling soil structure is not considered: the effect of roots. The proposed alternative approaches, the water retention curve and QST, use intact samples and could, therefore, be more appropriate for determining certain aspects of soil structure. The mesoporosity and, by extension, macroaggregation are better correlated with the QST parameters than the WSA fraction. Since the mesoporosity is a good measure for preferential flow, the QST could better predict the preferential flow of water and associated solutes. This would be interesting in, for example, nutrient availability studies. Also, the other soil properties are well-correlated with the QST, making it a viable alternative for the WSA method that is less time- and labour-intensive. However, the variability on QST replicates was very high, which counteracts its ease of use. Follow-up studies using the QST to predict soil functional properties, such as water infiltration depths, non-equilibrium flow and nutrient bioavailability in intact soil, are necessary to validate these results.

CRedit authorship contribution statement

Toon van Dael: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Quinten Verwerft:** Writing – original draft, Investigation, Formal analysis. **Erik Smolders:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was funded by the European Union. However, the views and opinions expressed are those of the authors only and do not necessarily reflect those of the European Union or the European

Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible. This work is supported by an ERC grant (EXPOSOIL, 101054917).

During the preparation of this work, the author(s) used Grammarly in order to improve their writing style. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2025.117281>.

Data availability

Data will be made available on request.

References

- Agassi, M., Morin, J., Shainberg, I., 1985. Effect of Raindrop Impact Energy and water salinity on Infiltration rates of Sodic Soils. *Soil Science Society of America* 49 (1).
- Barber, S.A., 1995. *Soil nutrient bioavailability: a Mechanistic approach*, 2nd ed. John Wiley & Sons, Inc.
- Benjamin, J.G., Mikha, M.M., Vigil, M.F., 2008. Organic Carbon Effects on Soil Physical and Hydraulic Properties in a Semiarid Climate. *Soil Science Society of America Journal* 72 (5), 1357–1362. <https://doi.org/10.2136/sssaj2007.0389>.
- Bless, A.E.S., Colin, F., Crabit, A., Follain, S., 2022. Soil Aggregate Stability in Salt-Affected Vineyards: Depth-Wise Variability Analysis. *Land* 11 (4). <https://doi.org/10.3390/land11040541>.
- Bronick, C.J., Lal, R., 2005. Soil structure and management: A review. In *Geoderma* (vol. 124 (1–2), 3–22. <https://doi.org/10.1016/j.geoderma.2004.03.005>.
- S. Brown Measures of Shape: Skewness and Kurtosis 2011 <http://www.tc3.edu/instruct/sbrown/stat/shape.htm>.
- Brubaker, S.C., Holzhey, C.S., Brasher, B.R., 1992. Estimating the Water-Dispersible Clay Content of Soils. *Soil Science Society of America* 56, 1227–1232.
- Buckingham, E., 1907. *Studies on the movement of soil moisture*. Washington Printing Office.
- Craze, B., Hamilton, G.J., 1991. Soil physical properties. *Soils: Their Properties and Management: A Soil Conservation Handbook for New South Wales*. Sydney University Press.
- de Dios Herrero, J., Colazo, J.C., Buschiazzi, D., Galantini, J., 2020. Influence of sand gradation on compaction of loess soils. *Soil and Tillage Research* 196. <https://doi.org/10.1016/j.still.2019.104414>.
- Duiker, S.W., Rhoton, F.E., Torrent, J., Smek, N.E., Lal, R., 2003. Iron (Hydr)Oxide Crystallinity Effects on Soil Aggregation. *Soil Science Society of America Journal* 67 (2), 606–611. <https://doi.org/10.2136/sssaj2003.6060>.
- Elliott, E.T., 1986. Aggregate Structure and Carbon, Nitrogen, and Phosphorus in Native and Cultivated Soils. *Soil Science Society of America* 50, 627–633.
- Feng, X., Simpson, A.J., Simpson, M.J., 2005. Chemical and mineralogical controls on humic acid sorption to clay mineral surfaces. *Organic Geochemistry* 36 (11), 1553–1566. <https://doi.org/10.1016/j.orggeochem.2005.06.008>.
- Ghosh, T., Maity, P. P., Rabbi, S. M. F., Das, T. K., & Bhattacharyya, R. (2023). Application of X-ray computed tomography in soil and plant -a review. In *Frontiers in Environmental Science* (Vol. 11). Frontiers Media SA. doi: 10.3389/fenvs.2023.1216630.
- ISO 13320:2020 Particle Size Analysis — Laser Diffraction Methods (2020).
- ISO 23470. (2007). *Determination of effective cation exchange capacity (CEC) and exchangeable cations using a hexaminecobalt trichloride solution*.
- Jarvis, N.J., 2007. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *European Journal of Soil Science* 71 (3), 279–302. <https://doi.org/10.1111/ejss.12973>.
- Johannes, A., Sauzet, O., Matter, A., Boivin, P., 2023. Soil organic carbon content and soil structure quality of clayey cropland soils: A large-scale study in the Swiss Jura region. *Soil Use and Management* 39 (2), 707–716. <https://doi.org/10.1111/sum.12879>.
- Kleber, M., Sollins, P., Sutton, R., 2007. A conceptual model of organo-mineral interactions in soils: Self-assembly of organic molecular fragments into zonal structures on mineral surfaces. *Biogeochemistry* 85 (1), 9–24. <https://doi.org/10.1007/s10533-007-9103-5>.
- Kubienna, W.L., 1938. *Micropedology*. Collegiate Press Inc.
- Kumaragamage, D., Warren, J., Spiers, G., 2021. Soil Chemistry. In *Digging into Canadian Soils*.
- Le Bissonnais, Y., 1996. Aggregate stability and assessment of soil crustability and erodibility: I. Theory and methodology. *European Journal of Soil Science* 67 (1), 11–21. <https://doi.org/10.1111/ejss.4.12311>.
- Methods of Testing Soils for Engineering Purposes - Soil Classification Tests - Dispersion - Determination of the Percent Dispersion of a Soil (1980).
- Newman, A.C.D., 1984. The significance of clays in agriculture and soils. *Philosophical Transactions of the Royal Society* 311, 375–389.
- Oades, J.M., 1993. The role of biology in the formation, stabilization and degradation of soil structure. In *Geoderma* 56.
- Oades, J.M., Waters, A.C., 1991. Aggregate Hierarchy in Soils. In *Aust. J. Soil Res* 29.
- Panagea, I.S., Apostolakis, A., Berti, A., Bussell, J., Cermak, P., Diels, J., Elsen, A., Kusá, H., Piccoli, I., Poesen, J., Stoate, C., Tits, M., Toth, Z., Wyseure, G., 2022. Impact of agricultural management on soil aggregates and associated organic carbon fractions: analysis of long-term experiments in Europe. *Soil* 8 (2), 621–644. <https://doi.org/10.5194/soil-8-621-2022>.
- Papadopoulos, A., Bird, N.R.A., Whitmore, A.P., Mooney, S.J., 2009. Investigating the effects of organic and conventional management on soil aggregate stability using X-ray computed tomography. *European Journal of Soil Science* 60 (3), 360–368. <https://doi.org/10.1111/j.1365-2389.2009.01126.x>.
- Rabot, E., Wiesmeier, M., Schlüter, S., Vogel, H.-J., Rabot, E., Wiesmeier, M., Schlüter, S., Vogel, H.-J., 2018. Soil structure as an indicator of soil functions: A review Soil structure as an indicator of soil functions: A review Soil structure as an indicator of soil functions: A review. *Geoderma* 314. <https://doi.org/10.1016/j.geoderma.2017.11.009i>.
- Regelink, I.C., Stoof, C.R., Rousseva, S., Weng, L., Lair, G.J., Kram, P., Nikolaidis, N.P., Kercheva, M., Banwart, S., Comans, R.N.J., 2015. Geoderma Linkages between aggregate formation, porosity and soil chemical properties. *Geoderma* 247–248, 24–37. <https://doi.org/10.1016/j.geoderma.2015.01.022>.
- Richie, J.A., 1963. Earthwork tunneling and the application of the soil testing procedures. *Journal of Soil Conservation* 19, 111–129.
- Safar, F., Whalen, J.K., 2023. Mechanical stability of newly-formed soil macroaggregates influenced by calcium concentration and the calcium counter-anion. *Geoderma* 430. <https://doi.org/10.1016/j.geoderma.2023.116333>.
- Schwertmann, U., 1991. Solubility and dissolution of iron-oxides. *Plant Soil* 130, 1–25. <https://doi.org/10.1007/BF00011851> Published: JAN 1991.
- Six, J., Elliott, E.T., Paustian, K., 2000. Soil Structure and Soil Organic Matter II. A Normalized Stability Index and the Effect of Mineralogy. *Soil Science Society of America Journal* 64 (3), 1042–1049. <https://doi.org/10.2136/sssaj2000.6431042x>.
- Stătescu, F., Zaucă, D.C., Pavel, L.V., 2013. Soil structure and water-stable aggregates. *Environmental Engineering and Management Journal* 12 (4), 741–746. <https://doi.org/10.30638/eemj.2013.091>.
- The Royal Society. (2020). *Soil structure and its benefits*.
- Totsche, K. U., Amelung, W., Gerzabek, M. H., Guggenberger, G., Klumpp, E., Knief, C., Lehdorff, E., Mikutta, R., Peth, S., Prechtel, A., Ray, N., & Kögel-Knabner, I. (2018). Microaggregates in soils. In *Journal of Plant Nutrition and Soil Science* (Vol. 181, Issue 1, pp. 104–136). Wiley-VCH Verlag. doi: 10.1002/jpln.201600451.
- USDA. (2008). *Soil Quality Indicators: Slaking*.
- Vanwindekens, F.M., Hardy, B.F., 2023. The QuantiSlakeTest, measuring soil structural stability by dynamic weighing of undisturbed samples immersed in water. *SOIL* 9 (2), 573–591. <https://doi.org/10.5194/soil-9-573-2023>.
- Weil, R. R., & Brady, N. C. (2016). *The nature and properties of soils* (D. Fox, Ed.; 15th ed.). Weng, L., Van Riemsdijk, W.H., Hiemstra, T., 2008. Humic nanoparticles at the oxide-water interface: Interactions with phosphate ion adsorption. *Environmental Science and Technology* 42 (23), 8747–8752. <https://doi.org/10.1021/es801631d>.
- Xiao, H., Liu, G., Zhang, Q., Fenli, Z., Zhang, X., Liu, P., Zhang, J., Hu, F., Elbasit, M.A.M. A., 2018. Quantifying contributions of slaking and mechanical breakdown of soil aggregates to splash erosion for different soils from the Loess plateau of China. *Soil and Tillage Research* 178, 150–158. <https://doi.org/10.1016/j.still.2017.12.026>.
- Xiao, L., Yao, K., Li, P., Liu, Y., Chang, E., Zhang, Y., Zhu, T., 2020. Increased soil aggregate stability is strongly correlated with root and soil properties along a gradient of secondary succession on the Loess Plateau. *Ecological Engineering* 143. <https://doi.org/10.1016/j.ecoleng.2019.105671>.
- Zhang, Y., Cao, Z., Hou, F., Cheng, J., 2021. Characterizing preferential flow paths in texturally similar soils under different land uses by combining drainage and dye-staining methods. *Water (Switzerland)* 13 (2). <https://doi.org/10.3390/w13020219>.
- Zhang, Y., Niu, J., Yu, X., Zhu, W., Du, X., 2015. Effects of fine root length density and root biomass on soil preferential flow in forest ecosystems. *Forest Systems* 24 (1), 1–13.